

Learning Objectives

1. Explain the digestion and absorption of dietary lipids
2. Illustrate the function of the chylomicron pathway
3. Explain the causes and consequences of fat malabsorption
4. Compare and contrast the life cycle of the various lipoprotein particles (VLDL, IDL, LDL, and HDL) with respect to their functions, composition, metabolism and transport.
5. Describe the clinical relevance related to dietary lipid absorption

The structure of plasma lipoproteins - I

- Both **triglyceride** (or triacylglycerol) and **cholesterol** may be absorbed from diet or synthesized in the body (de novo). They cannot be **transported** as free molecules due to their nonpolar nature
- Triglyceride and cholesterol are transported in plasma as **lipoproteins** particles— complexes of lipids and proteins (apoproteins). Lipoproteins are mixed **micelles** consisting of a core of nonpolar triglycerides and cholesterol (esters) surrounded by a surface layer of phospholipids and apoproteins.

Triglyceride (TG) or
Triacylglycerol (TAG)

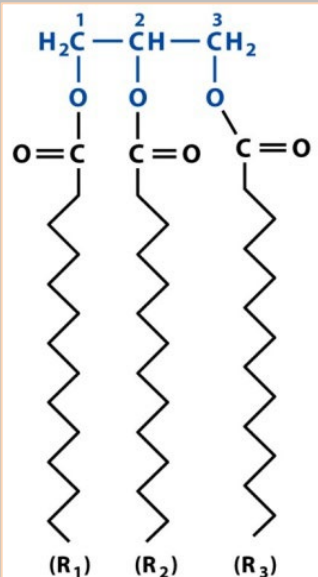
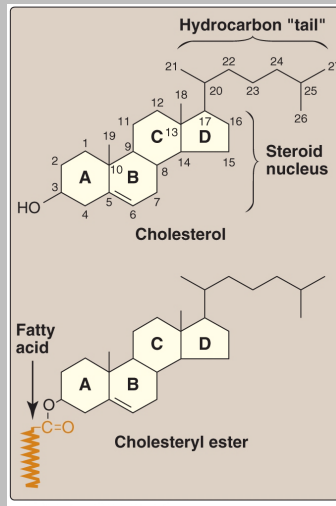
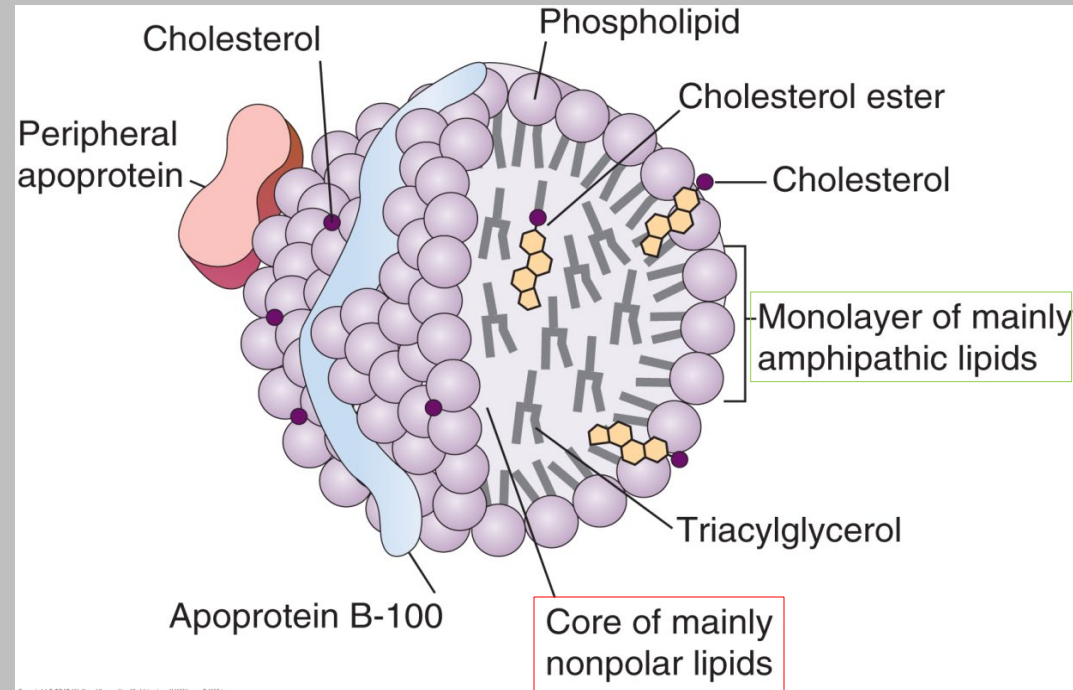


Figure 9-5b Principles of Biochemistry, 4/e
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Cholesterol

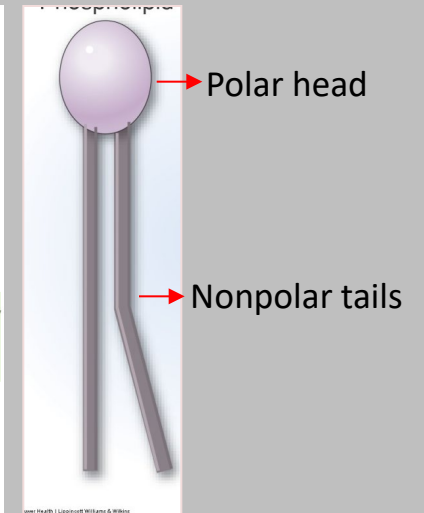


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Phospholipid



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A summary of the functions of lipoproteins

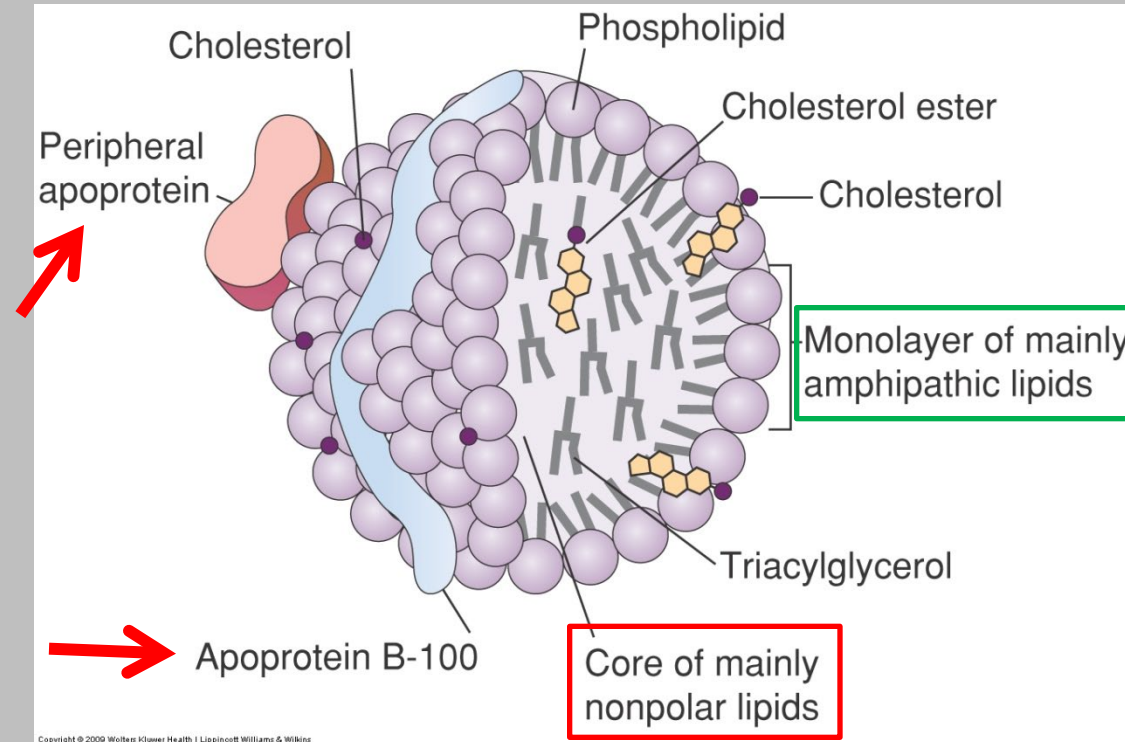
Functions:

- “**Solubilize**” the insoluble lipids for transport in the blood
- **Transport** both dietary lipids and endogenously synthesized lipids

Classification:

- **Chylomicrons** transport **dietary lipids** and **fat-soluble vitamins** from intestine to liver and other tissues
- **VLDLs** transport **endogenous synthesized lipids** from liver to other tissues
- **LDLs** deliver **cholesterol** from liver to tissues with more cholesterol needs (adrenals, sex glands, etc.)
- **HDLs** transport **cholesterol** from peripheral tissues back to the liver

Structural features of plasma lipoproteins -apoproteins



Chylomicron
VLDL
IDL
LDL
HDL

- **Shell:** phospholipids and other amphipathic molecules on the surface
- **Core** contains lipids (triglyceride and cholesterol esters)
- **Apoproteins** (also amphipathic):
 - Act as structural components (e.g., apoB-48, apoB-100, apoA)
 - Regulate enzyme activity (e.g., apoA-I, apoC-II)
 - Act as ligands for receptors (e.g., apoB, apoE)

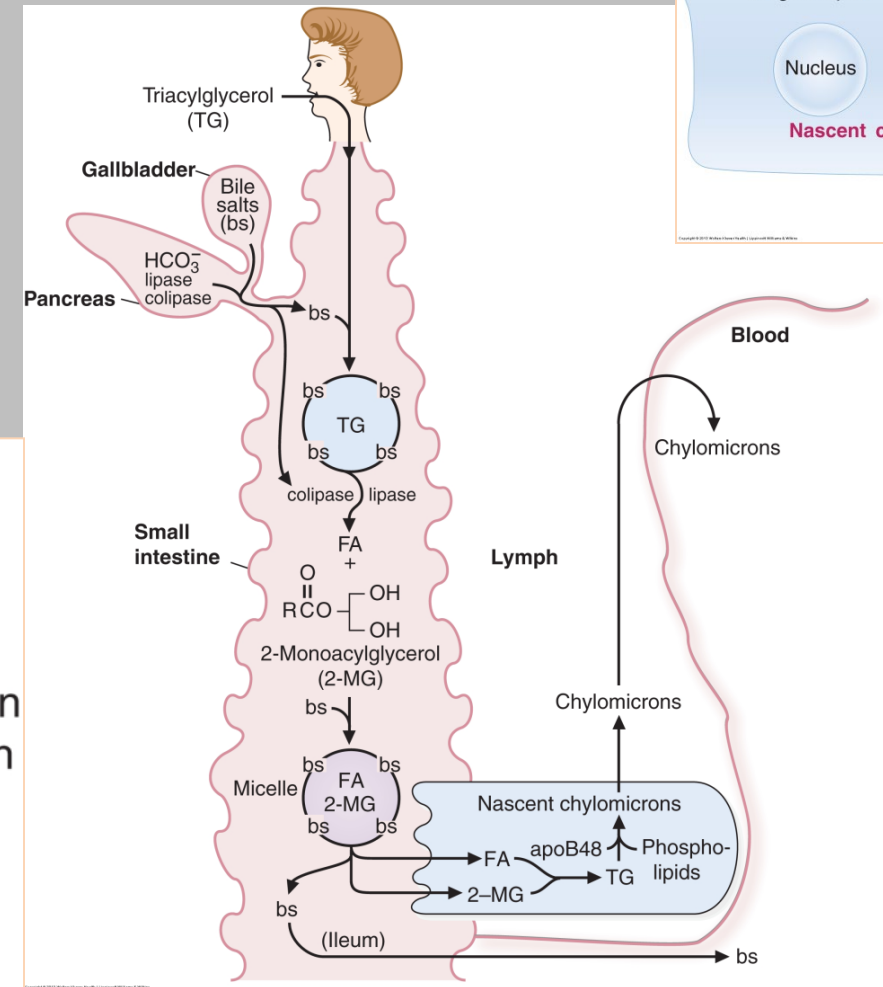
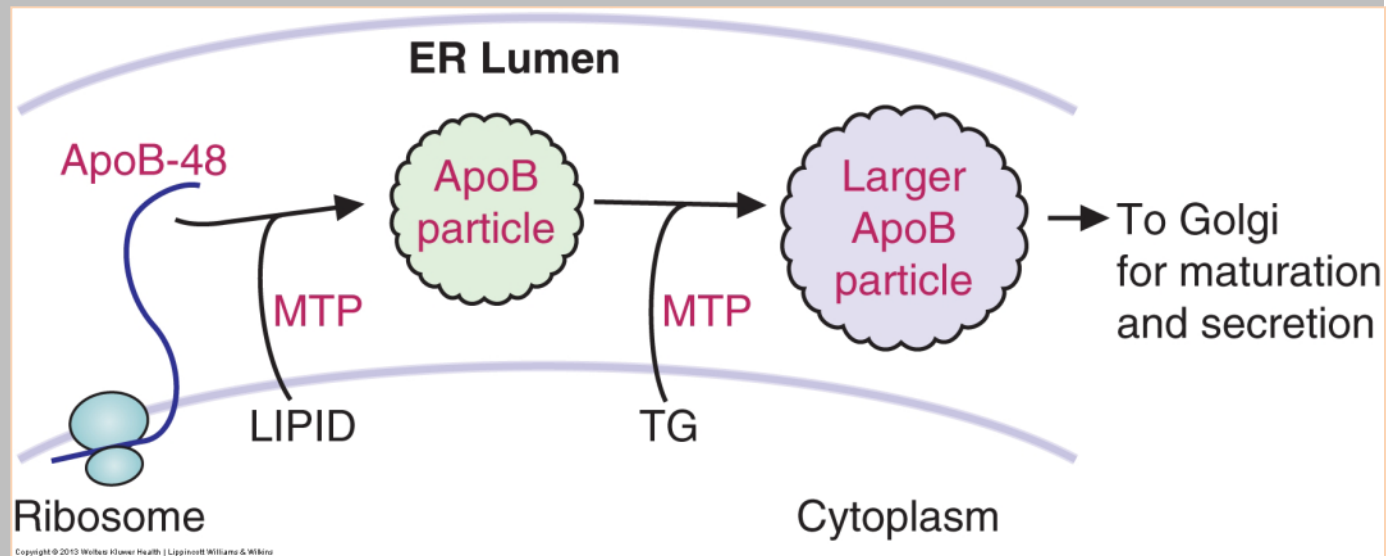
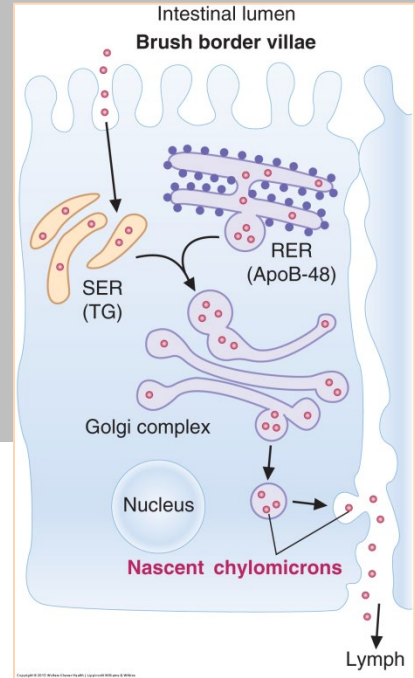
The three major pathways of lipoprotein metabolism

- The **chylomicron/chylomicron remnant pathway**: distributes **dietary triglycerides** and **cholesterol** to **liver** and peripheral tissues
- The **very low density lipoproteins (VLDL)/intermediate density lipoproteins (IDL)/low density lipoproteins (LDL) pathway**: transports **triglycerides** and **cholesterol** out of **liver** to the peripheral tissues
- The **nascent/mature high density lipoprotein (HDL) pathway**: transports **cholesterol** from peripheral tissues back to **liver**

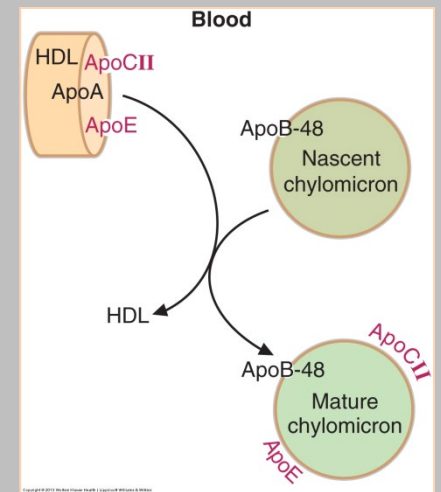
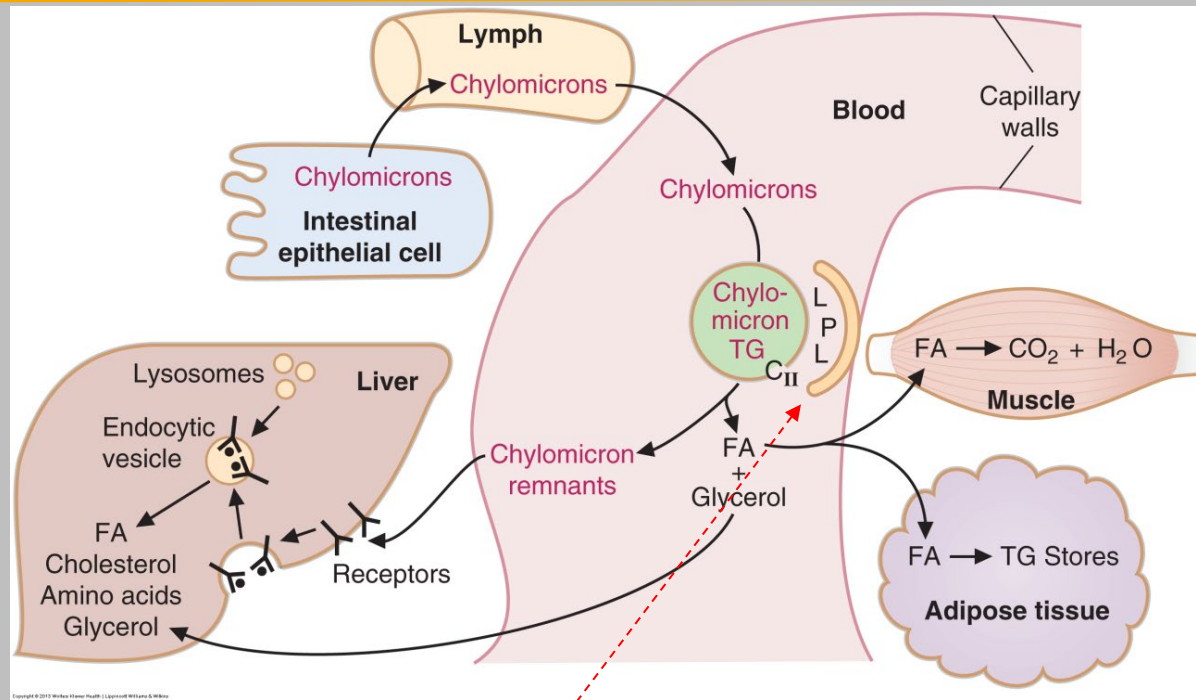
Transport of dietary lipids by chylomicrons - synthesis

Chylomicrons: synthesized in the **intestinal epithelial cells**:

- **ApoB-48** is produced in ER (next slide).
- Lipids are loaded onto **ApoB-48** by **MTP** (or MTTP, **Microsomal triglyceride Transfer Protein**).
- Defect in MTP causes **Abetalipoproteinemia**
- Secreted into the **lymphatic system**
- Enter the blood through the **thoracic duct**



Transport of dietary lipids by chylomicrons: maturation and fate



Maturation - transfer of apoproteins from HDLs to the nascent chylomicrons:

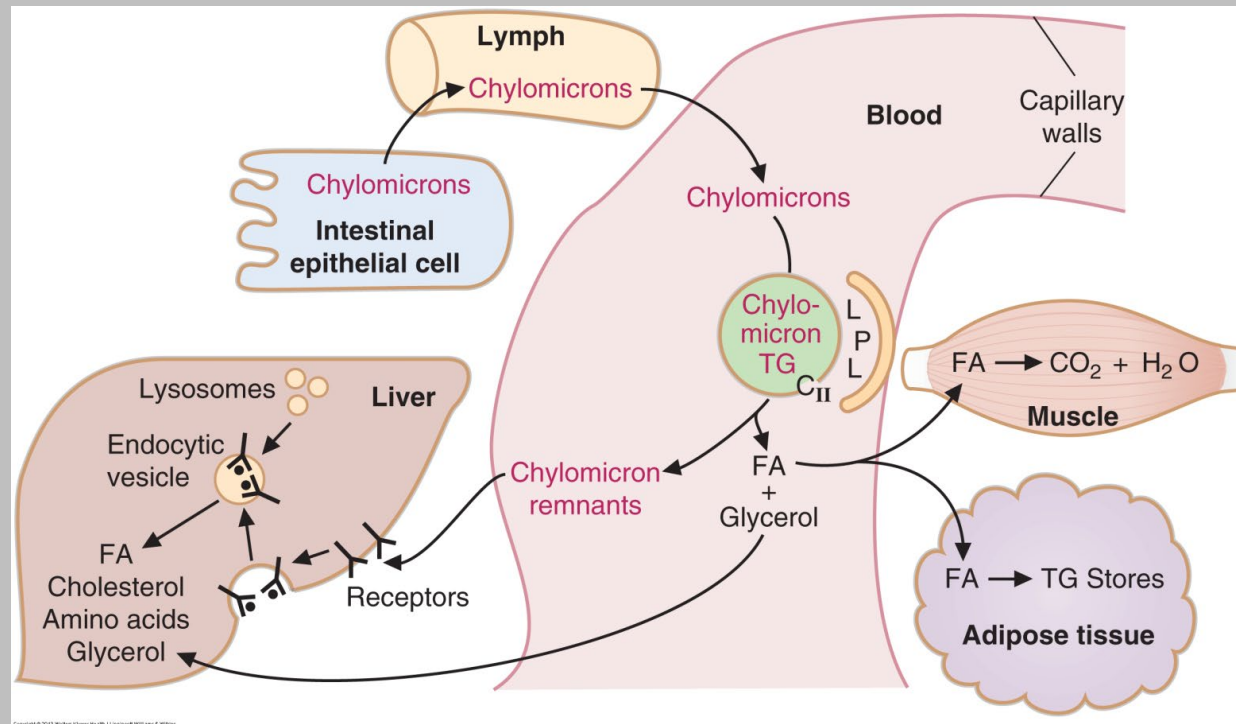
- **ApoC-II** and **ApoE** are transferred to the nascent chylomicrons to form mature chylomicrons
 - **ApoC-II** stimulates LPL
 - **ApoE** recognized by receptors

Fate - **Lipoprotein lipase (LPL)** hydrolyzes the TGs in the chylomicrons:

- LPL is attached to proteoglycans on the **basement membrane** of capillary endothelium
- LPL is high in adipose tissues, muscle, and mammary gland
 - **Insulin** stimulates LPL synthesis in adipose tissues after a balanced meal
- LPL is activated by **ApoC-II**
 - Released fatty acids may enter the cell or bind albumin in blood
 - glycerol used by the liver for lipid synthesis, glycolysis, or gluconeogenesis
- The remaining chylomicron is thus called **chylomicron remnant**, which can be taken up by liver through binding to the receptors on the surface of hepatocytes and endocytosis

Transport of dietary lipids by Chylomicrons - fate

- **Chylomicron remnants** decrease in size and increase in density after lose of TGs
- **ApoC-II** returns to HDLs
- **Chylomicron remnants** can be taken up by the hepatocytes
 - Bound to the hepatocyte **apoE receptor** via **apoE**
 - Taken into hepatocytes through endocytosis
 - The endocytic vesicles subsequently fuse with lysosomes
 - The contents are hydrolyzed and used by liver for biosynthesis





Clinical relevance to the absorption of dietary lipids

Fat malabsorption

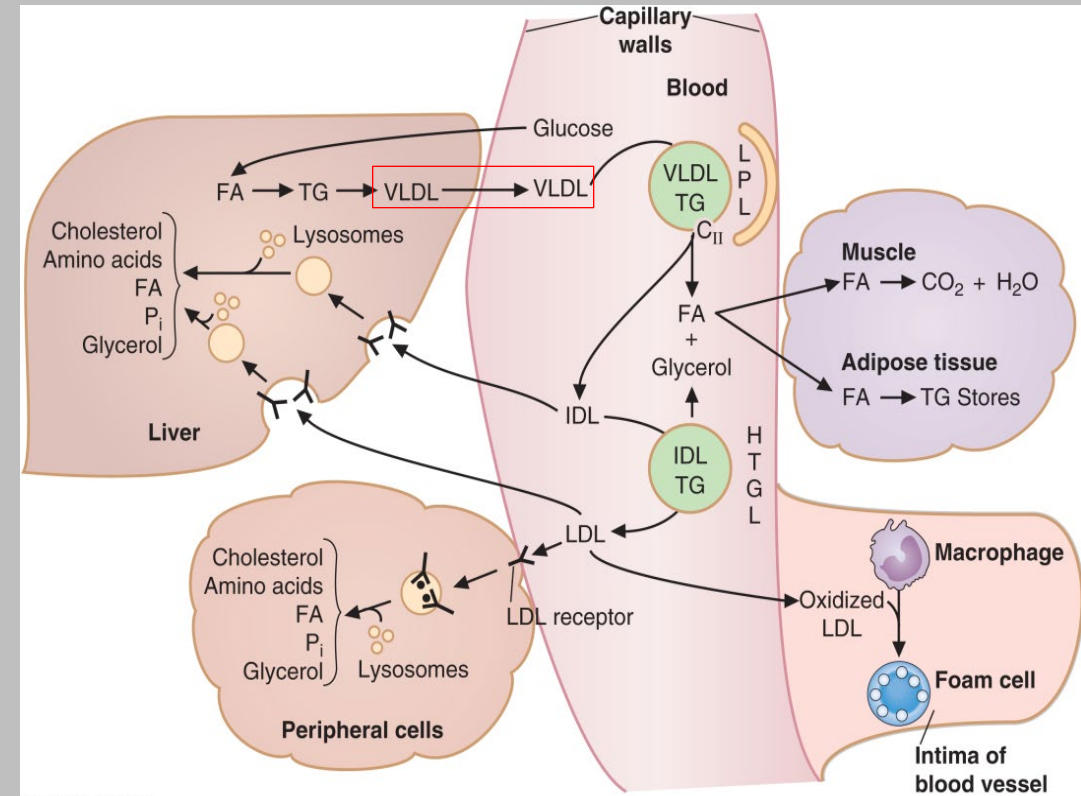
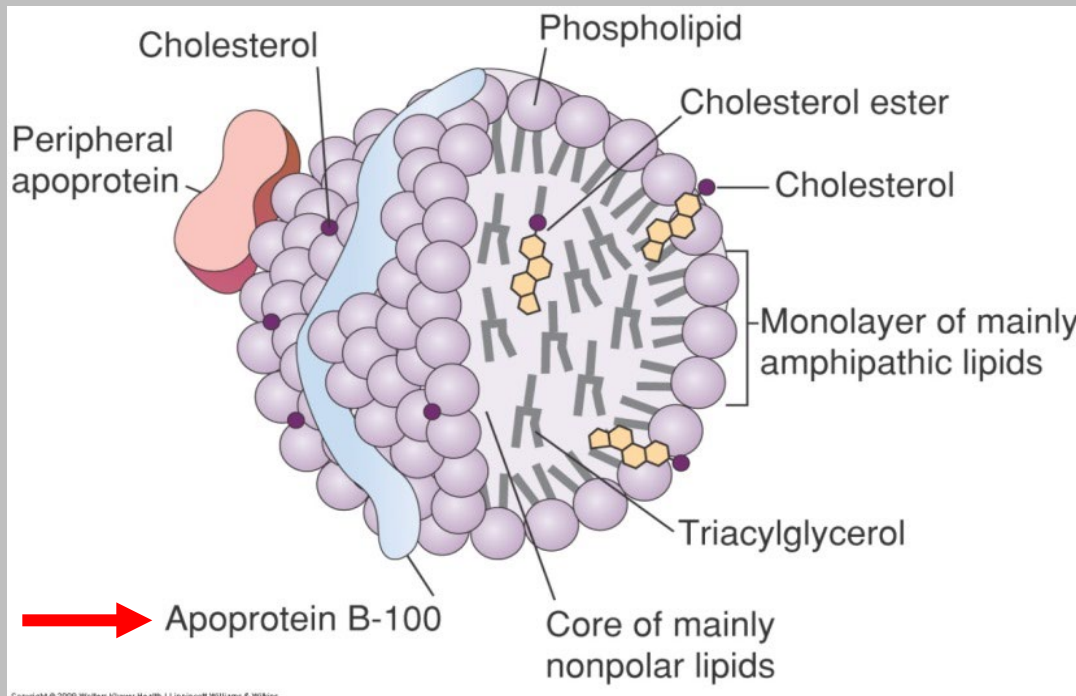
- Possible causes:
 - Pancreatic failure (pancreas secretes a variety of **digestive enzymes** including lipase and colipase**)
 - Lack of **bile salts** (bile duct blockage)
 - Intestinal diseases
- Malabsorption causes **Steatorrhea** with bulky fatty appearing in stools
- Decrease in **fat soluble vitamin** absorption (D,E,A,K) because they are absorbed and transported via the Chylomicron pathway as well.

**Colipase:

- 1) Anchors lipase at the lipid-aqueous interface
- 2) Restores activity to lipase in the presence of inhibitory substances like bile salts that bind the micelles.

Transport of **endogenous** lipids by VLDL-I

- **VLDL** is synthesized in the **liver**
 - Secreted into the **hepatic veins** as **nascent VLDL**
 - Accept **apoE** and **apoC-II** from **HDL** in blood and become **mature VLDL**
 - Function to carry the lipids to peripheral tissues: muscle (used as energy), adipose (storage) etc.
- **ApoB-100** functions as the structural apoprotein for **VLDL** (next slide)
- **Microsomal triacylglycerol transfer proteins** (MTTP, or **MTP**) load lipids onto ApoB-100 to form **VLDL**
- **Lipoprotein lipase** (**LPL**) also digests the TGs of **VLDL**

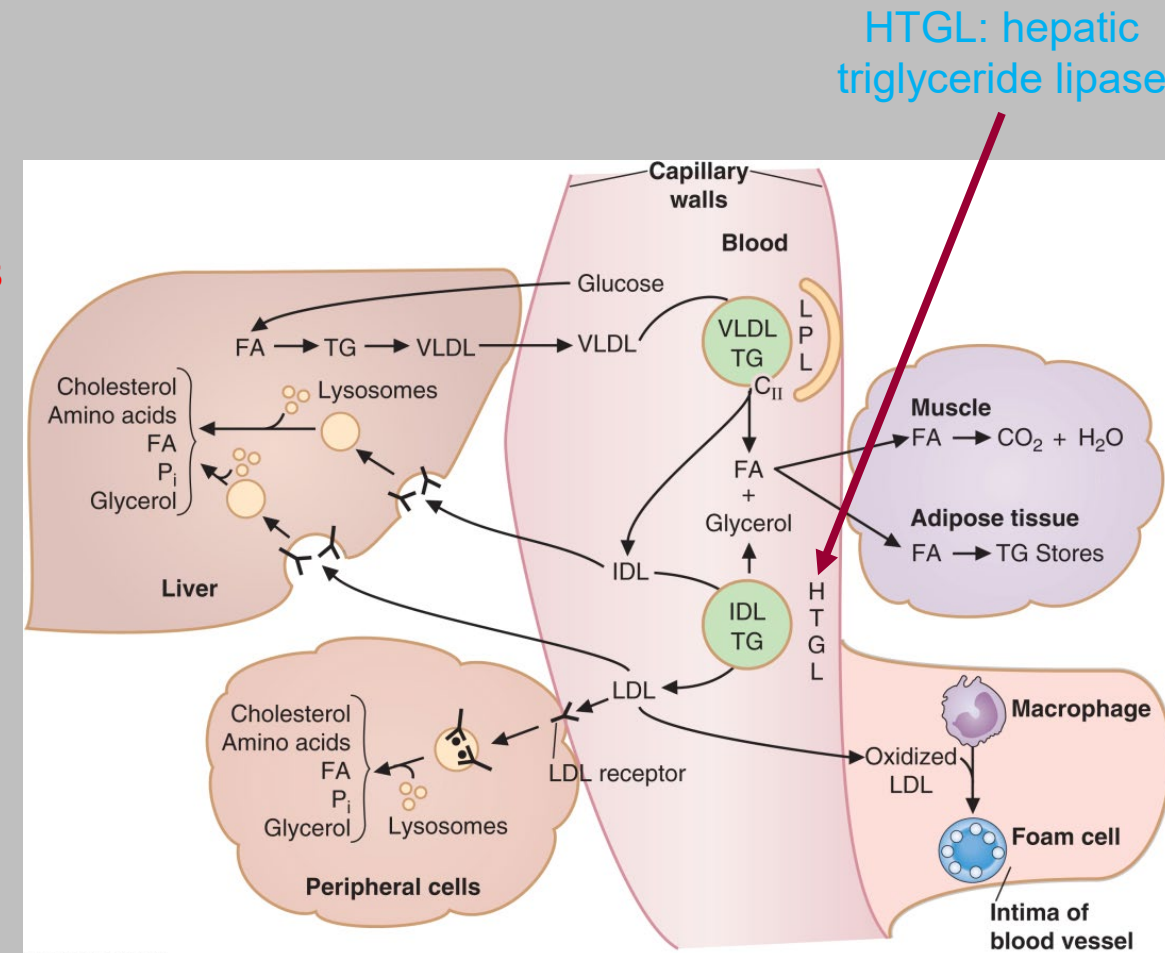
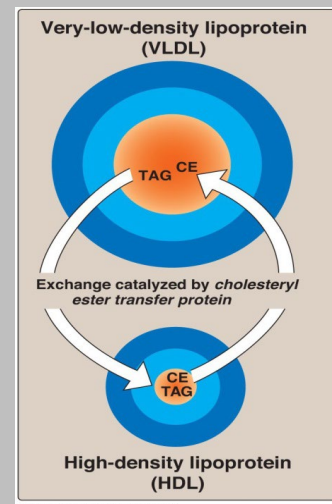


Transport of **endogenous** lipids by VLDL-II

VLDL becomes **VLDL remnant** or **IDL** after losing TG

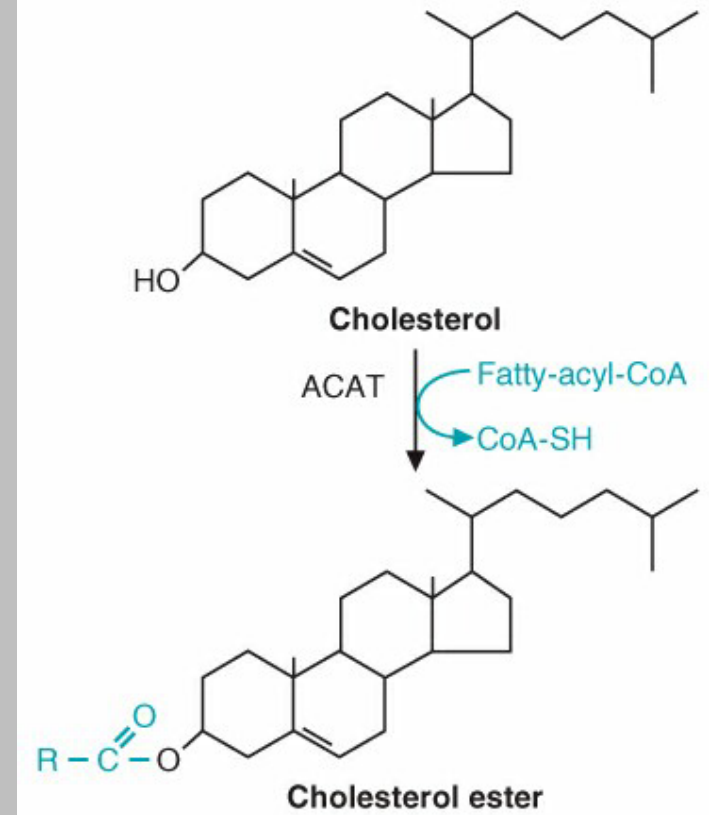
- **IDL** can be taken up by liver hepatocytes via **apoE receptor**
- **IDL** can also be further metabolized and become **LDL**
 - successive action of **LPL/HTGL** to further hydrolyze TGs
 - remaining **LDL** is rich of **cholesterol** (from **HDL**)

- **LDL** is taken up by liver or peripheral tissues
 - binds to the **LDL receptor** via **apoB-100**
 - **LDL** provides **cholesterol** to peripheral tissues
 - **high LDL levels can saturate the uptake by liver**
 - Excess and oxidized **LDL** taken up by **macrophages**
 - Forms **foam cells** then plaque - **atherosclerosis**



Transport of **endogenous** lipids by VLDL - III

- Inside the cell, free cholesterol can be re-esterified by Acyl CoA:Cholesterol AcylTransferase (**ACAT**)
 - Esterification keeps cholesterol out of the cytoplasmic membrane
- LDL receptor synthesis can be down-regulated by cholesterol
 - Involves **SREBP** (as discussed in the cholesterol synthesis lecture)
- **Niemann-Pick type C disease** cannot release cholesterol from the lysosome:
 - Cholesterol builds up inside **lysosomes**
 - Cells (nerve cells) die → early death



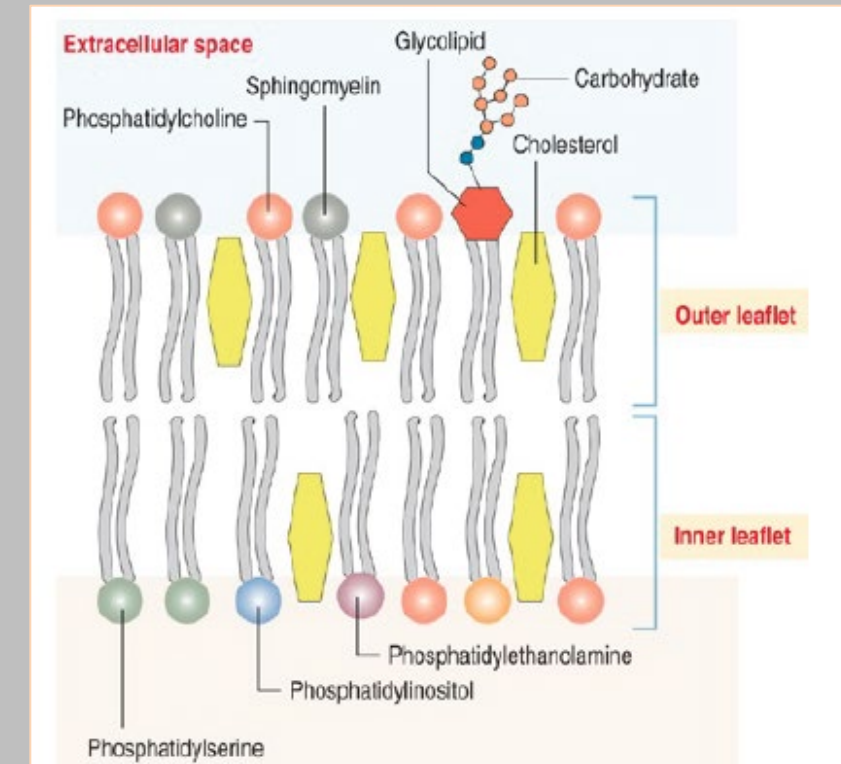


Cells/tissues get rid of their cholesterol by **reverse cholesterol transport:**

- **ABC1** (an ATPase, also known as ABCA1) flips cholesterol from the **inner leaflet** of plasma membrane to the **outer leaflet**
 - **Tangier disease**: ABCA1 deficiency → reduced HDL
- Cholesterol is then transferred to HDL

HDL picks up cholesterol from extrahepatic cells/tissues:

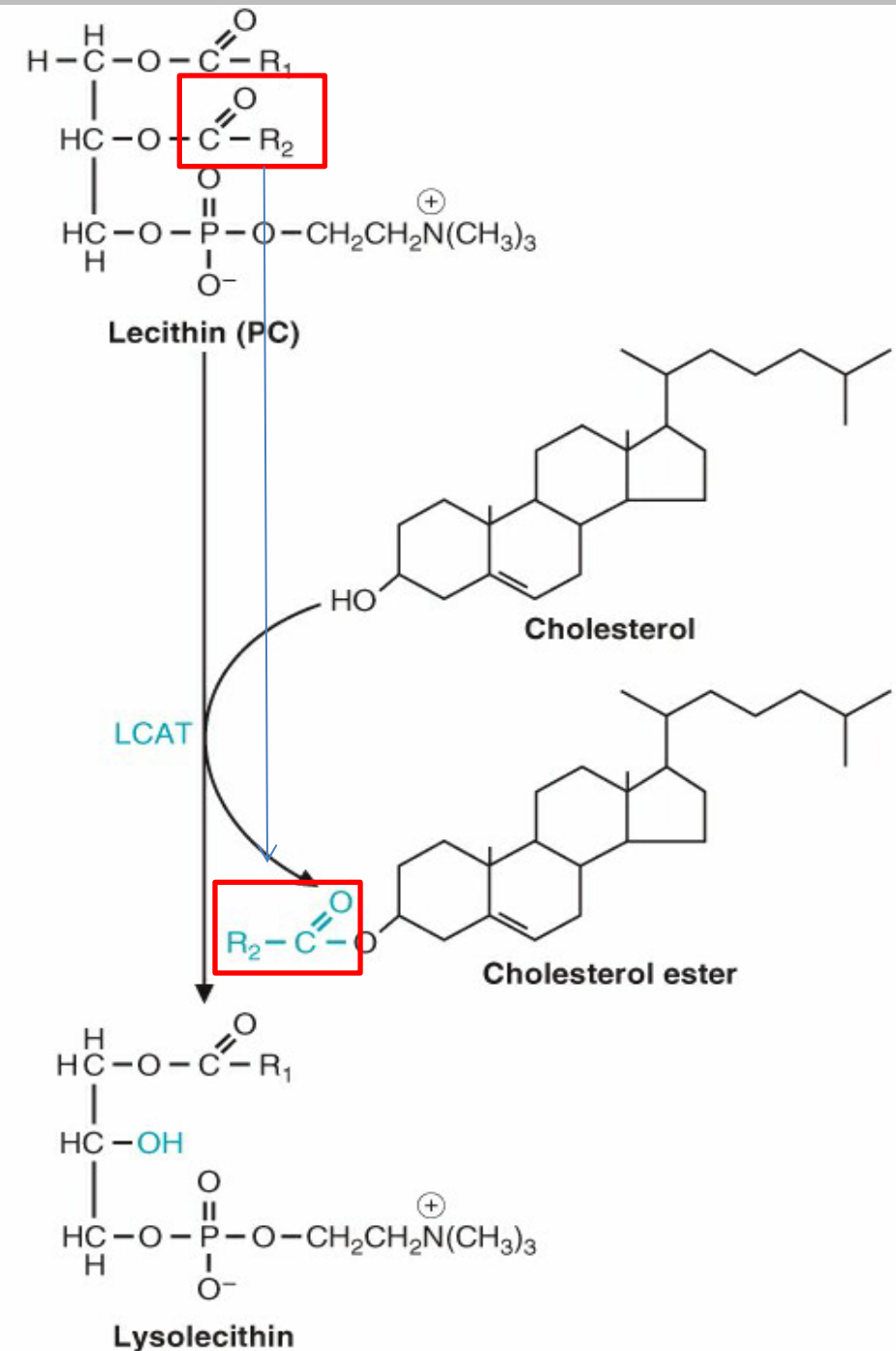
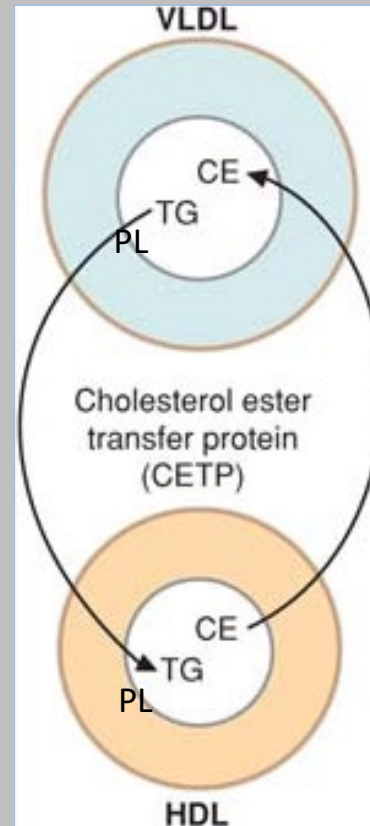
- Requires 2 enzymes:
 - LCAT** (aka **PCAT**-phosphatidylcholine) = Lecithin: cholesterol acyl transferase (activated by **ApoA-I**)
 - CETP** = Cholesterol ester transfer protein



HDL in reverse cholesterol transport-III

HDL picks up cholesterol from extrahepatic cells/tissues requires 2 enzymes (**LCAT** and **CETP**):

- **LCAT** - lecithin cholesterol acyl transferase
 - Activated by **ApoA-I**
 - synthesizes cholesterol ester (CE)
 - FA comes from Phosphatidylcholine (lecithin)
 - **traps cholesterol in the HDL**
- **CETP** - cholesterol ester (CE) transfer proteins
 - transfers cholesterol ester to VLDL in exchange for triglyceride (TG) and phospholipid (PL)



Summary of plasma lipoproteins

Chylomicrons

- Produced in **intestinal** epithelial cells from **dietary fat**
- Carries lipids in blood

VLDL

- Produced in **liver**
- Carries lipids in blood

IDL

- Produced in blood (**remnant of VLDL** after TG digestion)
- Endocytosed by liver or converted to LDL

LDL

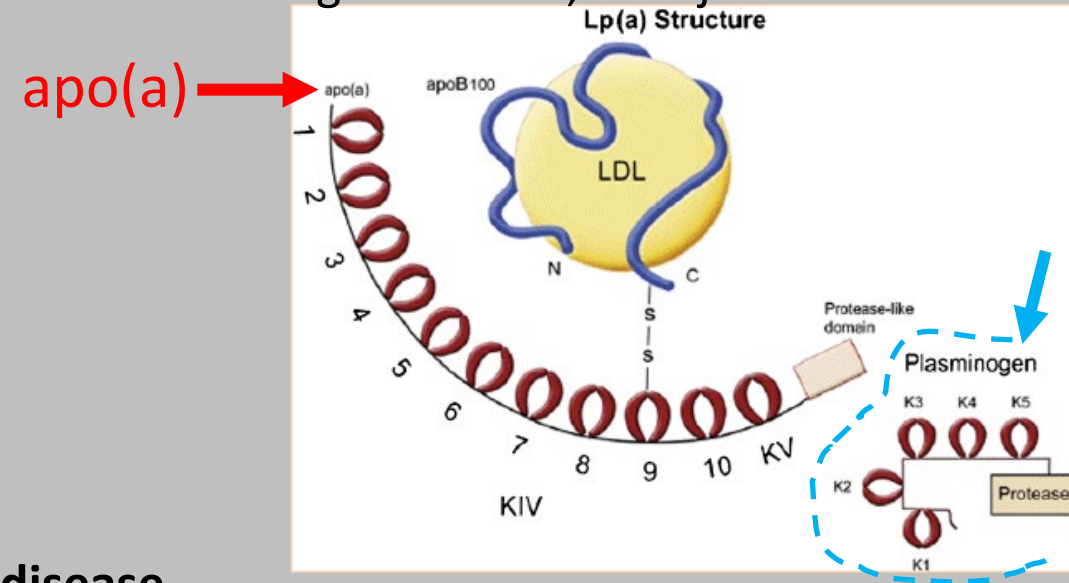
- Produced in blood (**remnant of IDL** after TG digestion; the end product of VLDL)
- Contain **high concentration of cholesterol and cholesterol ester**
- Endocytosed by liver and peripheral tissues via LDL receptors

HDL

- Produced in **liver** and **intestine**
- **Exchanges proteins and lipids** with other lipoproteins (e.g., chylomicrons, VLDL)
- Functions in the **return of cholesterol from peripheral tissues to liver, Reverse Cholesterol Transport**

Lp(a) - one last clinically important lipoprotein

- Lipoprotein(a), or Lp(a), is **nearly identical to LDL**. A distinguishing feature of Lp(a) is the presence of **apoprotein(a)**, or **apo(a)** that is covalently linked to a Cys residue on **apoB-100**
- **apo(a)** is structurally similar to **plasminogen** (a precursor of plasmin that degrade fibrin, a major component of blood clots.)



- High levels associated with increased risk of coronary heart disease
 - Levels mostly genetically controlled
 - High dietary **trans fats** are associated with increased level of Lp(a)
- The physiological function of Lp(a) remains unknown. One **hypothesis** is that elevated level of Lp(a) **slows the breakdown of blood clots** that trigger heart attack because it competes with **plasminogen** for binding of fibrin
- **Niacin** (vit. B3) reduces Lp(a), as well as LDL-cholesterol and TAG, and raises HDL

Summary

- **Key concepts:** chylomicrons, VLDL, IDL, LDL, HDL, Lp(a), apoproteins, MTP, Lipoprotein lipase (LPL), Reverse Cholesterol Transport
- Synthesis, maturation and fate of chylomicrons, VLDL, IDL, LDL, HDL
- Biological functions of chylomicrons, VLDL, IDL, LDL, HDL
- **Clinical correlations:** disease symptoms, biochemical basis, and treatment (Abetalipoproteinemia, fat malabsorption, Steatorrhea, Niemann-Pick type C disease, Tangier disease)